

01_Python Workflow & Dataset Audit

Objective

The objective of the Python phase was to prepare the Olist Brazilian E-Commerce dataset for advanced analytics by performing data inspection, quality assessment, validation, cleaning, feature engineering, exploratory analysis, and database integration.

Python served as the data processing layer of the project and ensured that only validated, analysis-ready datasets were used for SQL analysis, KPI development, and Power BI reporting.

Python Workflow

The following workflow was followed during the data processing phase:

Raw CSV Files

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Data Inspection

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Data Quality Assessment

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Referential Integrity Validation

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Data Cleaning & Preprocessing

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Feature Engineering

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Python Exploratory Data Analysis (EDA)

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PostgreSQL Integration

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SQL Analysis

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Power BI Dashboard Development

Dataset Inventory

The Olist marketplace dataset consists of multiple related tables representing customers, orders, products, payments, reviews, sellers, and geographic information.

Table Name	Rows
Customers	99,441
Geolocation	1,000,163
Orders	99,441
Order Items	112,650
Payments	103,886
Reviews	99,224
Products	32,951
Sellers	3,095
Category Translation	71

Data Quality Audit

An initial dataset audit was conducted to understand the structure, quality, and analytical readiness of each table.

The audit included:

- Dataset structure review
- Row and column assessment
- Datatype inspection
- Missing value identification
- Duplicate analysis
- Primary key candidate identification

- Composite key candidate identification
- Relationship discovery across tables

Key Findings

The audit identified a well-structured relational dataset suitable for business analysis.

Major observations included:

- Strong relational structure across transactional and descriptive tables.
- Primary key candidates identified for core entities.
- Composite key candidates identified for transaction-level tables.
- Datetime conversion requirements identified in Orders, Order Items, and Reviews.
- Missing values identified in Orders, Reviews, and Products.
- No critical duplicate key issues detected.
- Dataset approved for data cleaning and preprocessing.

Outcome

The dataset audit established a clear understanding of the marketplace data model, data quality characteristics, and table relationships. The findings provided the foundation for referential integrity validation, data cleaning, feature engineering, and subsequent analytical phases of the project.